Homework 2

ENEE 132 — Basic Electronics I (Semiconductor Devices)
Due Date: 21 July 2026

Instructions:
Show all work clearly and express all final answers with correct units. Students must include their name and index number on all submitted solutions. Do not copy the questions. Provide the solutions only.

Question

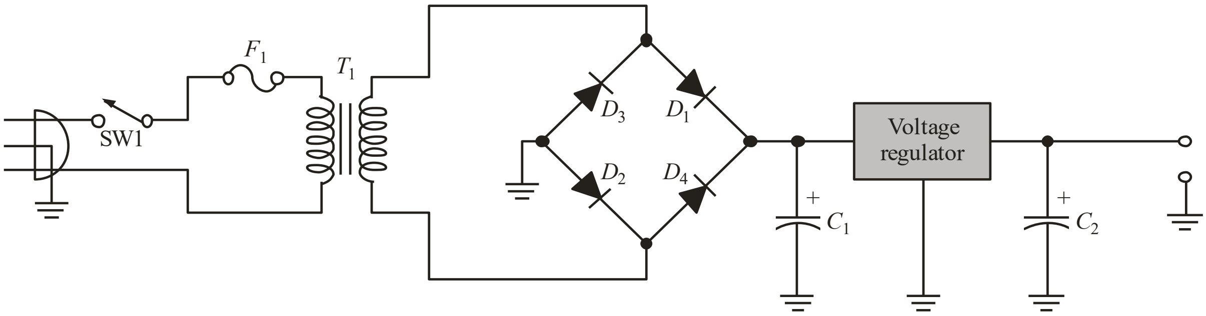

Figure 1

Refer to the regulated DC power supply in *Figure 1*, which is designed to deliver a 12 V output. The mains voltage is 230 V rms, 50 Hz. The transformer turns ratio is n = 0.06, the filter capacitor is C_1 = 4700 μF, and the bridge rectifier uses silicon diodes (assume the practical diode model). The design requires that the ripple factor at the filter output not exceed 3%.

(a) Calculate the peak secondary voltage

(b) Determine the peak rectified output voltage and sketch the corresponding full-wave rectified output waveform, clearly labelling its peak value and its period.

(c) Determine the minimum PIV rating required for each bridge diode, allowing a 20% safety margin.

(d) The regulator stage draws 200 mA from the filter output. Determine the load resistance presented to the filter, then calculate the ripple factor at the filter output and express it as a percentage. Confirm whether it satisfies the 3% design requirement.

(e) The regulator has a load regulation of 0.5%. If its full-load output voltage is 12.00 V, determine the no-load (unloaded) output voltage.

**** End of Assignment ****